

- Entropy Function
- Binary Cross Entropy Function (Cost, Loss)
- Multicategory Case
- Softmax Output
- Softmax Loss (Categorical Cross Entropy)

Principle of maximum entropy

$$H(x) = - \sum_i x_i \log(x_i)$$

$$H(x) = E(-\ln(p(x))) = - \int p(x) \ln(p(x)) dx$$

The **principle of maximum entropy** states that the **probability distribution** which best represents the current state of knowledge about a system is the one with largest **entropy**, in the context of precisely stated prior data (such as a **proposition** that expresses **testable information**).

https://en.wikipedia.org/wiki/Principle_of_maximum_entropy

Bernoulli

A Bernoulli random variable is simply a binary variable (i.e., it occupies one of two states) with

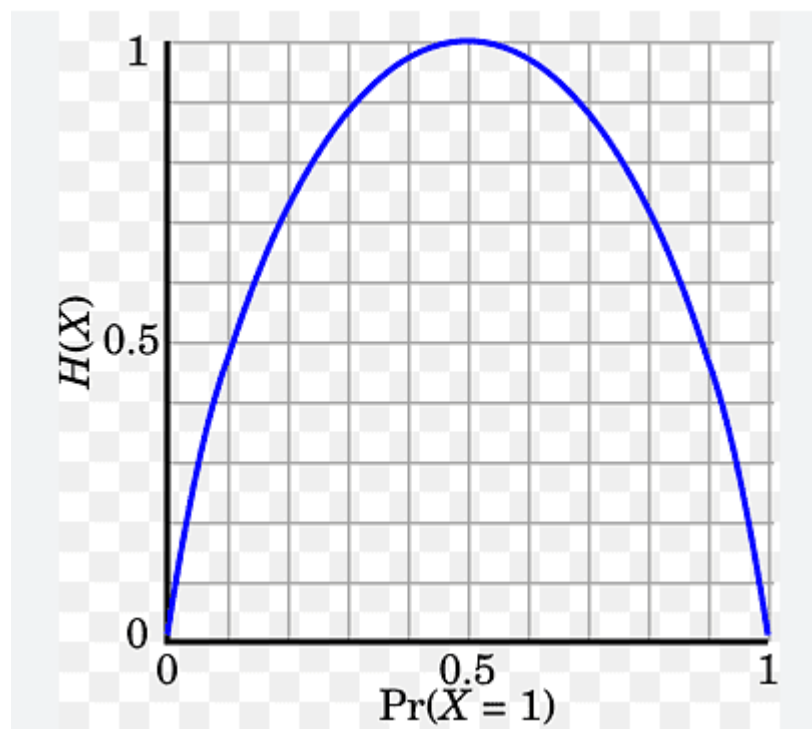
$$P(1) = p \quad (4)$$

$$P(0) = 1 - p \quad (5)$$

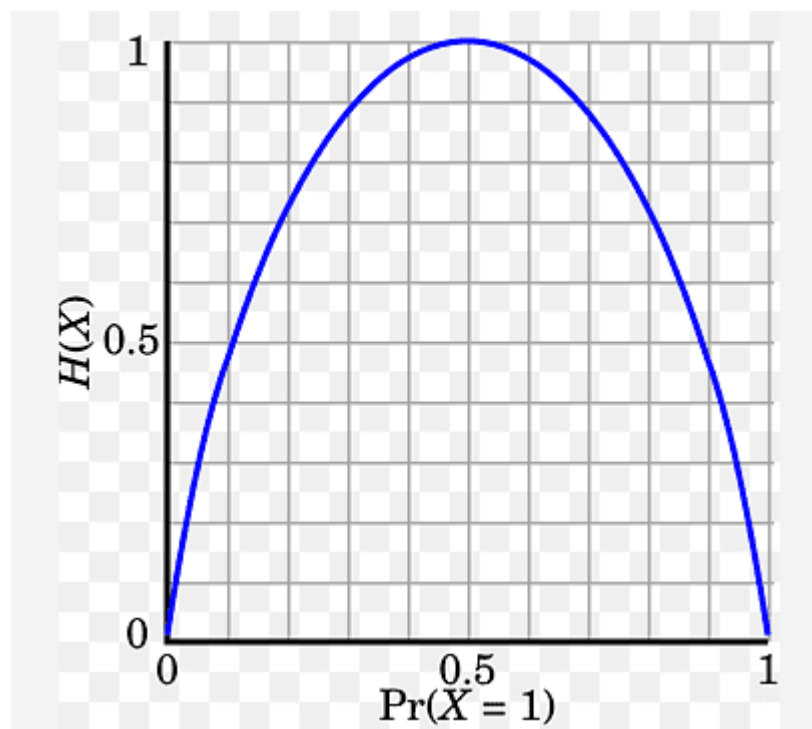
$$H(p) = -p \log(p) - (1 - p) \log(1 - p)$$

$$\max_p H(p)$$

$$H(p) = -p\log(p) - (1 - p)\log(1 - p)$$

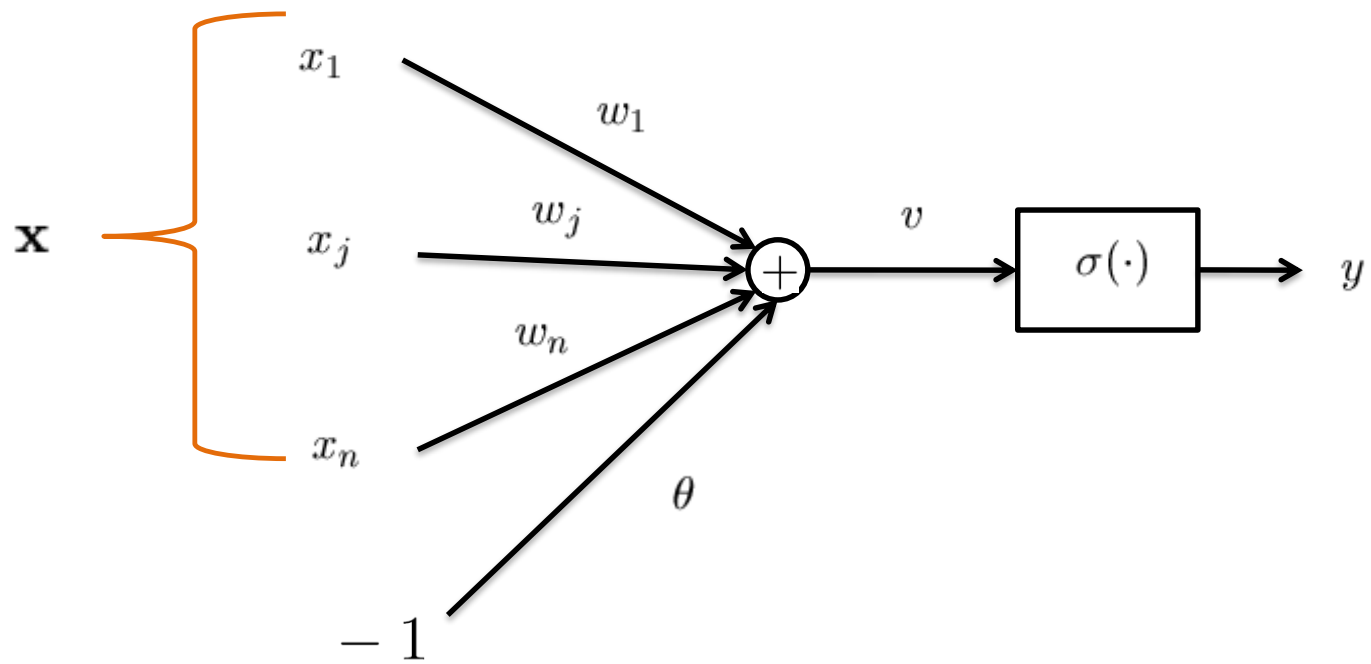


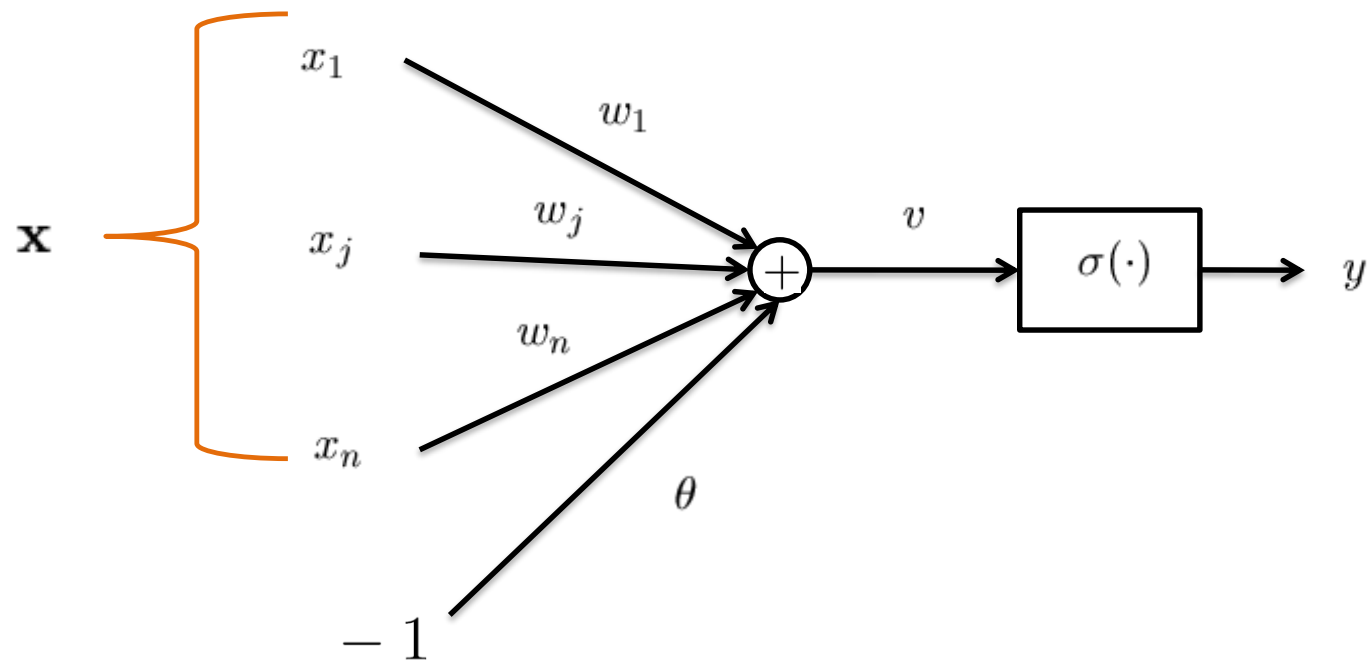
$$H(p) = -p \log(p) - (1-p) \log(1-p)$$



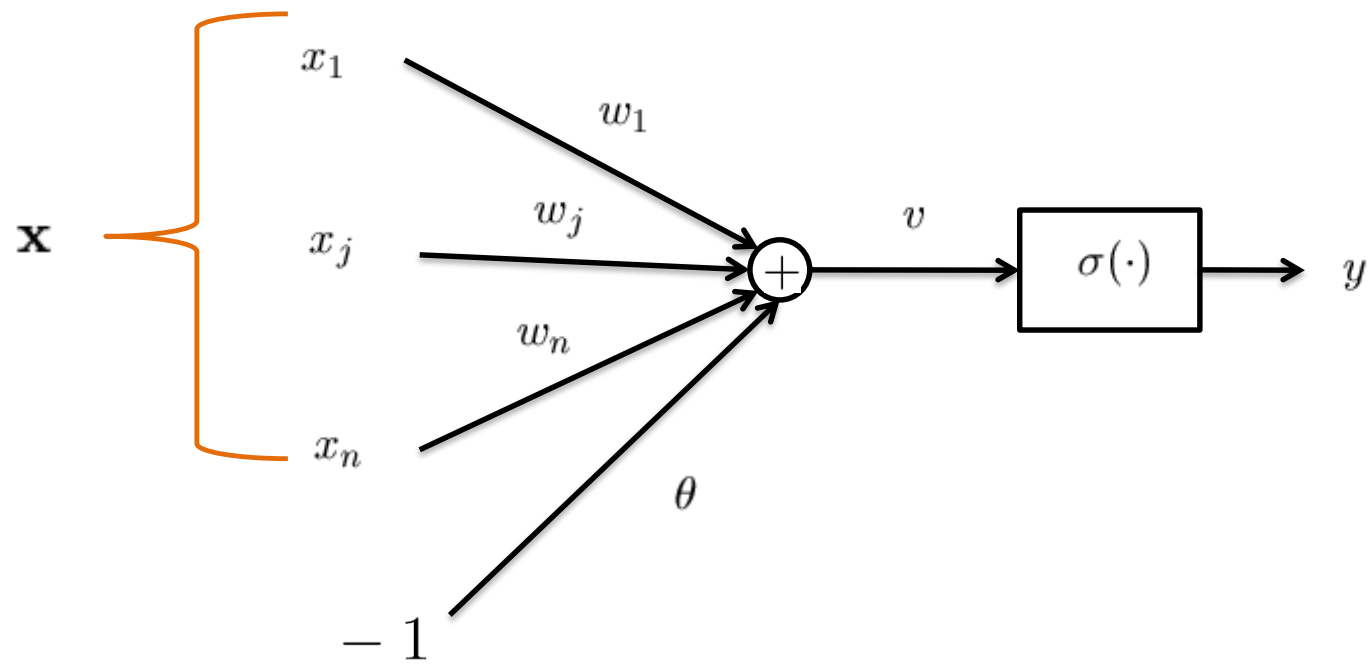
$$H(0) = 0 \quad , \quad H(1) = 0$$

$$H(p) > 0 \quad \text{for} \quad 0 < p < 1$$



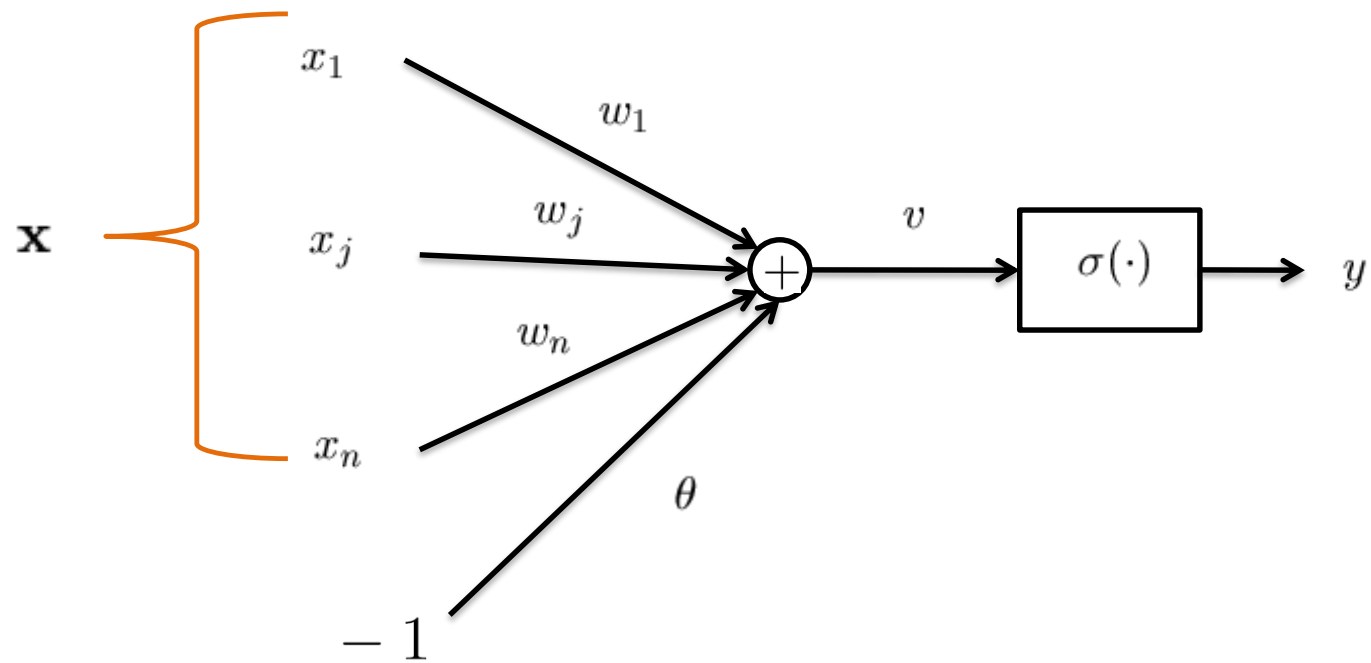


$\mathbf{x} \rightarrow d$  $E = 0.5(d - y)^2$



$$\min_{\mathbf{w}} E$$

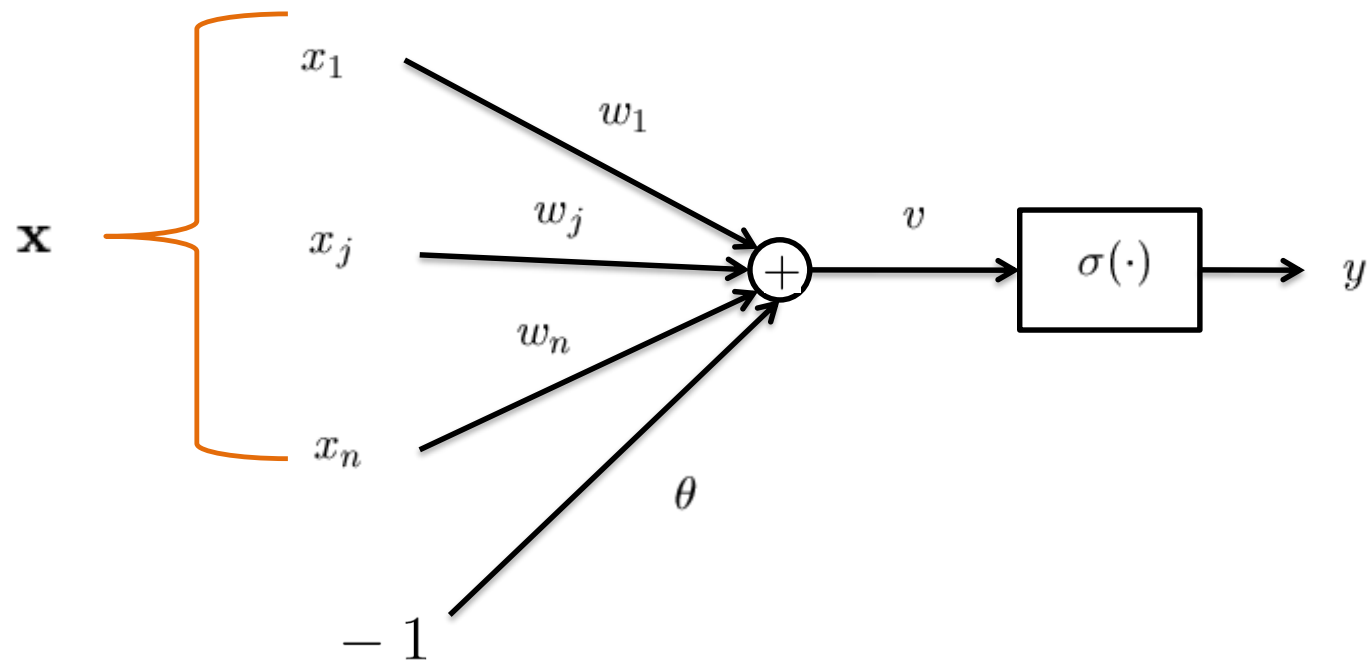
$$\mathbf{x} \rightarrow d \quad \longrightarrow \quad E = 0.5(d - y)^2$$



$$\min_{\mathbf{w}} E$$

$$w_j \leftarrow w_j - \eta \frac{\partial E}{\partial w_j}$$

$$\mathbf{x} \rightarrow d \quad \longrightarrow \quad E = 0.5(d - y)^2$$

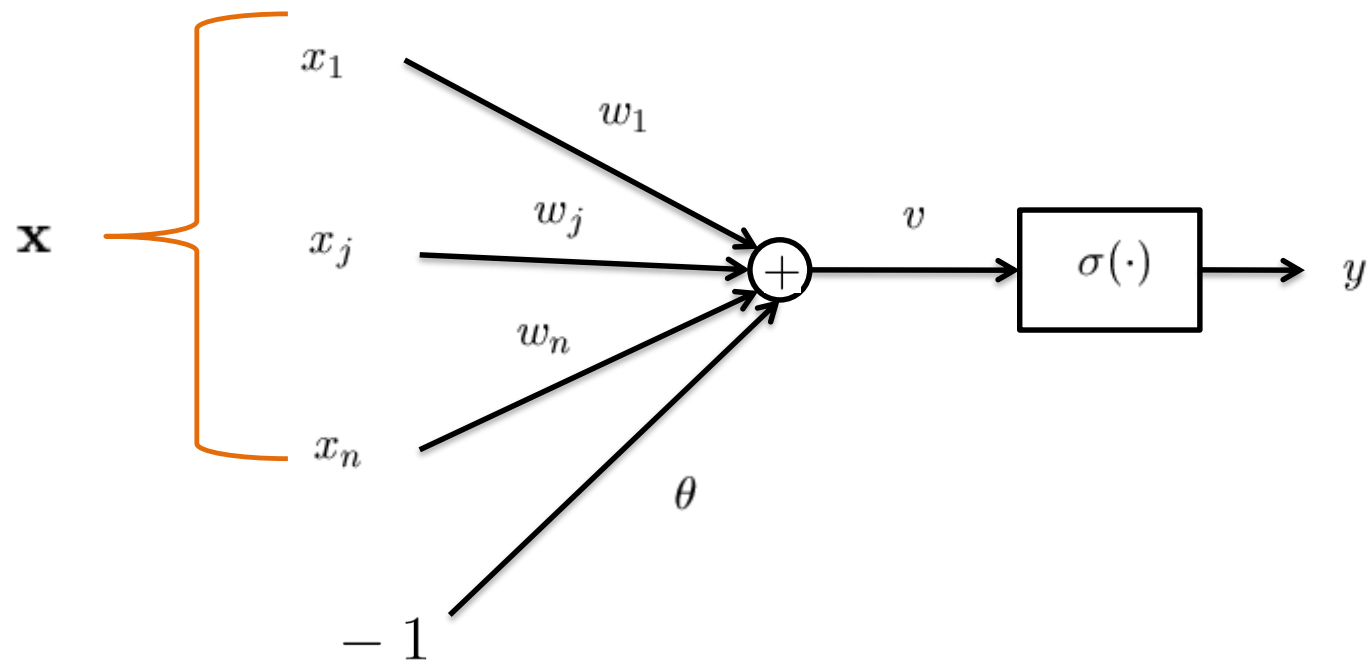


$$\min_{\mathbf{w}} E$$

$$w_j \leftarrow w_j - \eta \frac{\partial E}{\partial w_j}$$

$$\frac{\partial E}{\partial w_j} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j}$$

$$\mathbf{x} \rightarrow d \quad \longrightarrow \quad E = 0.5(d - y)^2$$



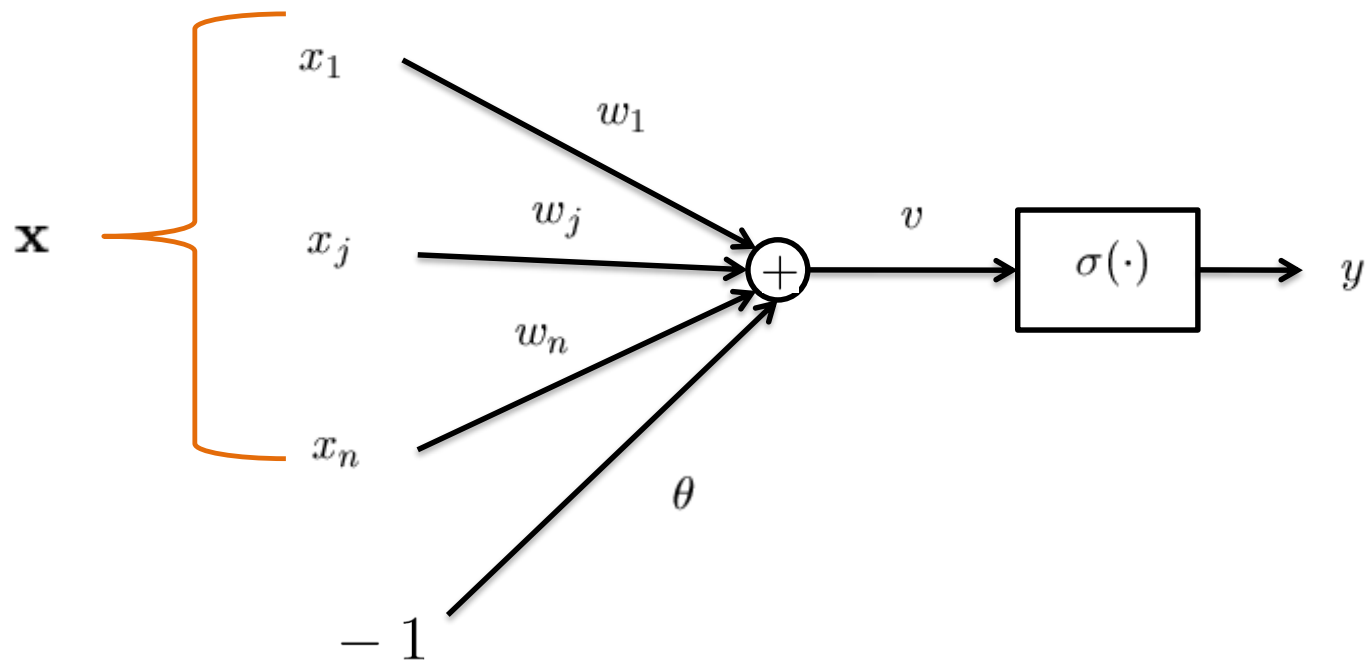
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$$\frac{\partial E}{\partial w_j} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j}$$

$$= -(d - y) \sigma'(v) x_j$$



$$\mathbf{x} \rightarrow d \quad \longrightarrow \quad E = 0.5(d - y)^2$$

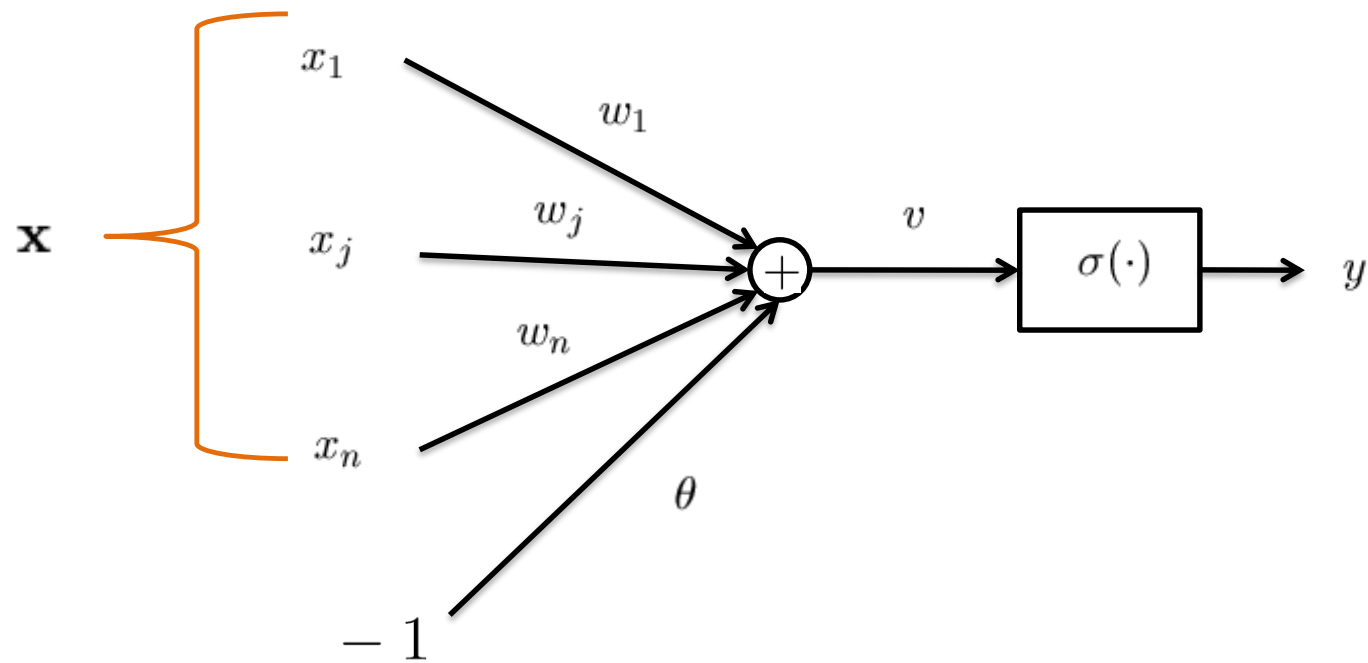
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$$\frac{\partial E}{\partial w_j} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j}$$

$$= -(d - y) \sigma'(v) x_j$$

$$= -(d - y) y (1 - y) x_j$$



$$\mathbf{x} \rightarrow d \quad \longrightarrow \quad E = 0.5(d - y)^2$$

$$w_j \leftarrow w_j + \eta(d - y)y(1 - y)x_j$$

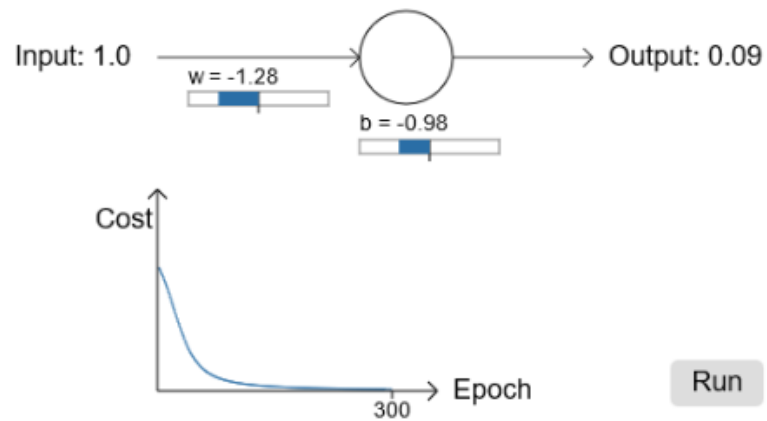
$$\min_{\mathbf{w}} E$$

$$w_j \leftarrow w_j - \eta \frac{\partial E}{\partial w_j}$$

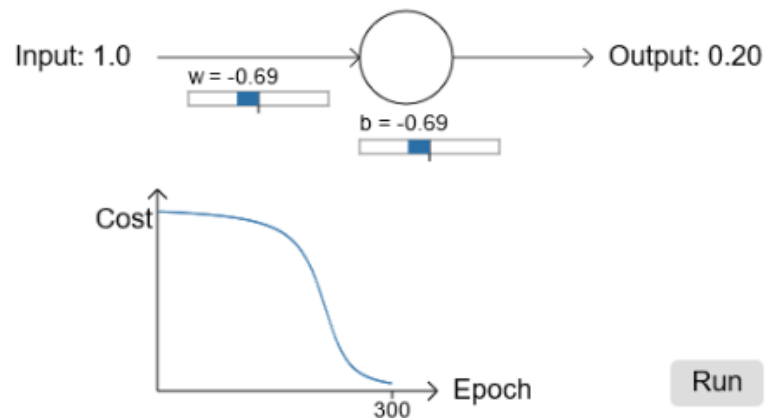
$$\frac{\partial E}{\partial w_j} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j}$$

$$= -(d - y)\sigma'(v)x_j$$

$$= -(d - y)y(1 - y)x_j$$



As you can see, the neuron rapidly learns a weight and bias that drives down the cost, and gives an output from the neuron of about 0.09. That's not quite the desired output, 0.0, but it is pretty good. Suppose, however, that we instead choose both the starting weight and the starting bias to be 2.0. In this case the initial output is 0.98, which is very badly wrong. Let's look at how the neuron learns to output 0 in this case. Click on "Run" again:



<http://neuralnetworksanddeeplearning.com/chap3.html>

2 Class Classification



Binary Classification

Class 1 $\rightarrow d$

Typically $d = 1$ or $d = 0$

Class 2 $\rightarrow 1 - d$

←

$$H(d) = -d \log(d) - (1-d) \log(1-d)$$



Entropy

2 Class Classification



Binary Classification

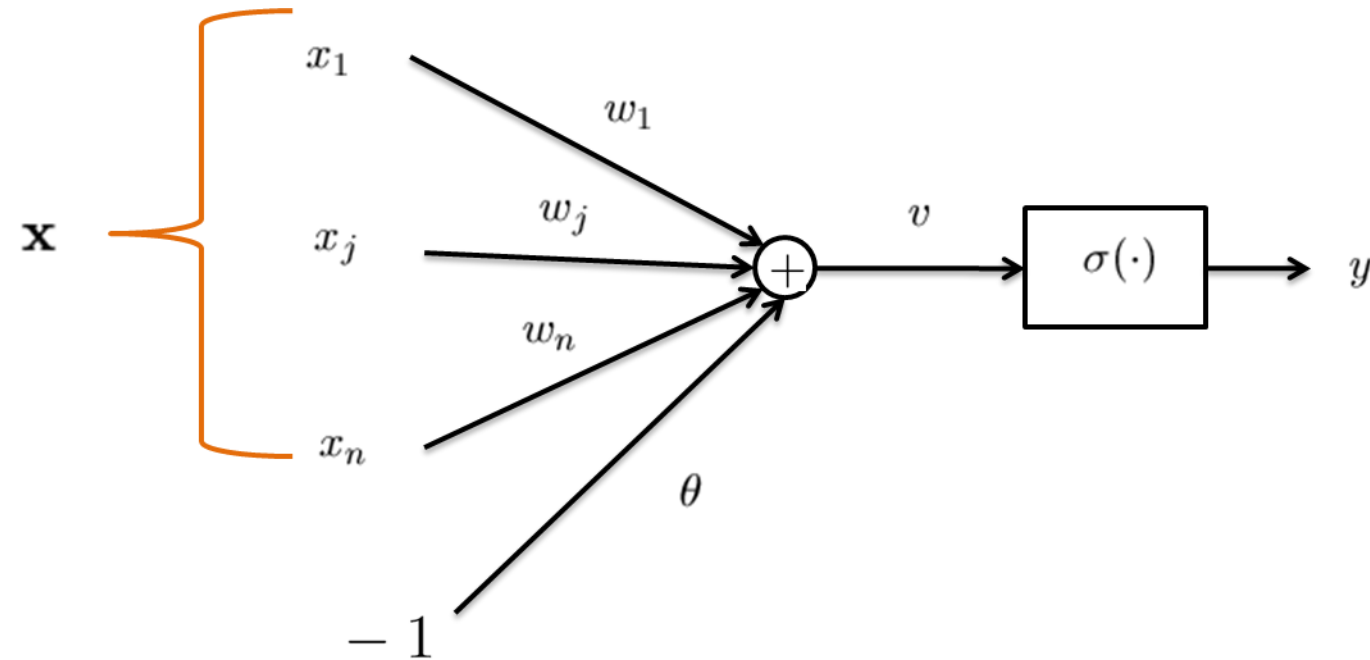
Class 1 $\rightarrow d$

Typically $d = 1$ or $d = 0$

$$H(d) = -d \log(d) - (1-d) \log(1-d)$$

Class 2 $\rightarrow 1 - d$

←



2 Class Classification



Binary Classification

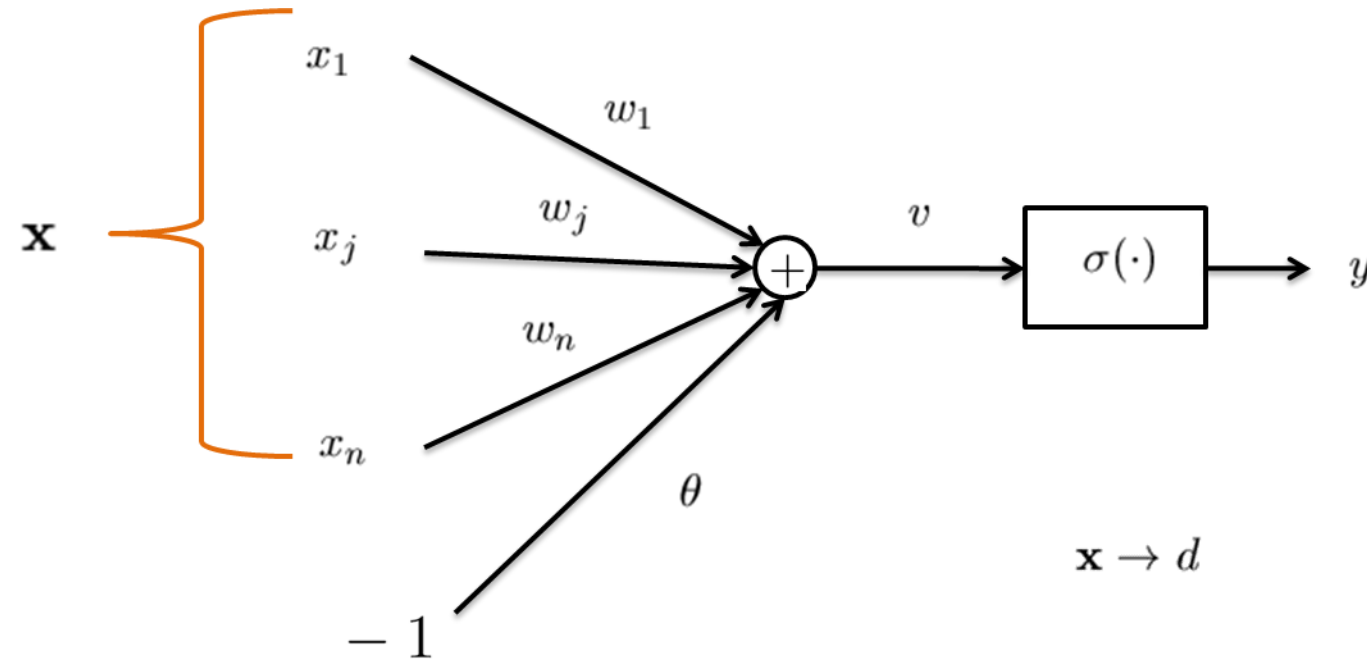
Class 1 $\rightarrow d$

Typically $d = 1$ or $d = 0$

$$H(d) = -d \log(d) - (1-d) \log(1-d)$$

Class 2 $\rightarrow 1 - d$

←



$$E = -d \log(y) - (1-d) \log(1-y)$$

2 Class Classification



Binary Classification

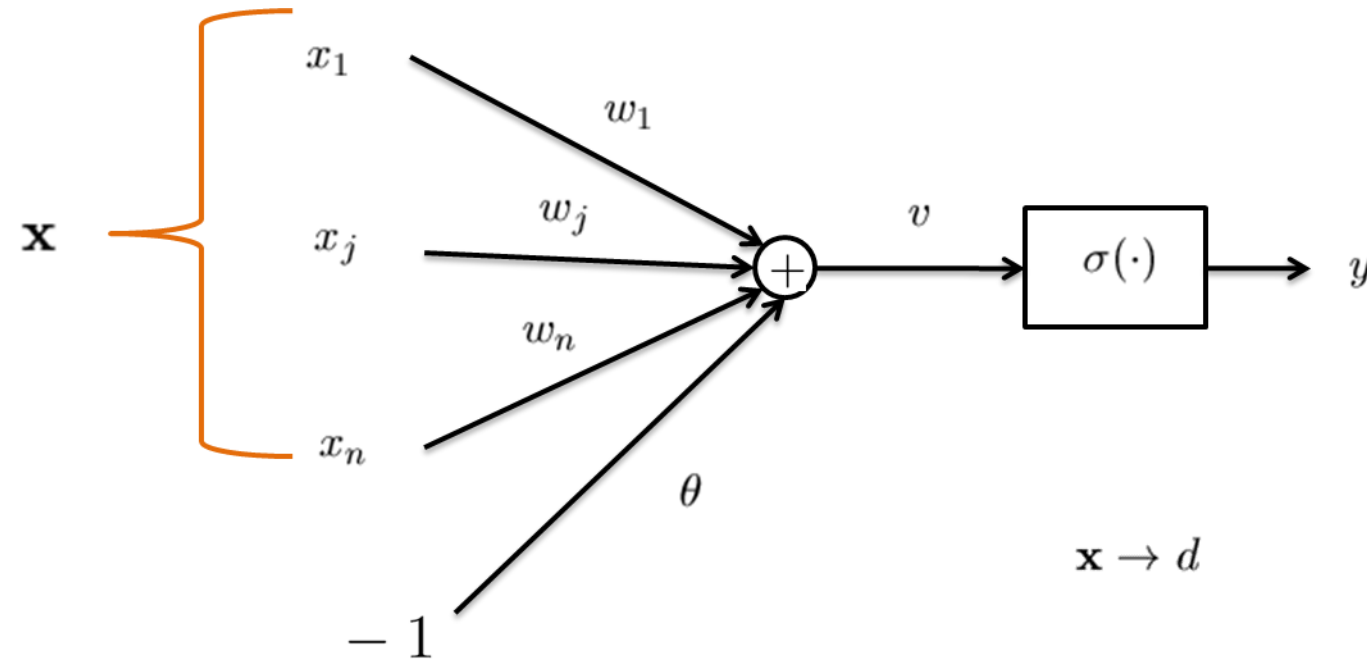
Class 1 $\rightarrow d$

Typically $d = 1$ or $d = 0$

$$H(d) = -d \log(d) - (1-d) \log(1-d)$$

Class 2 $\rightarrow 1 - d$

$\leftarrow$



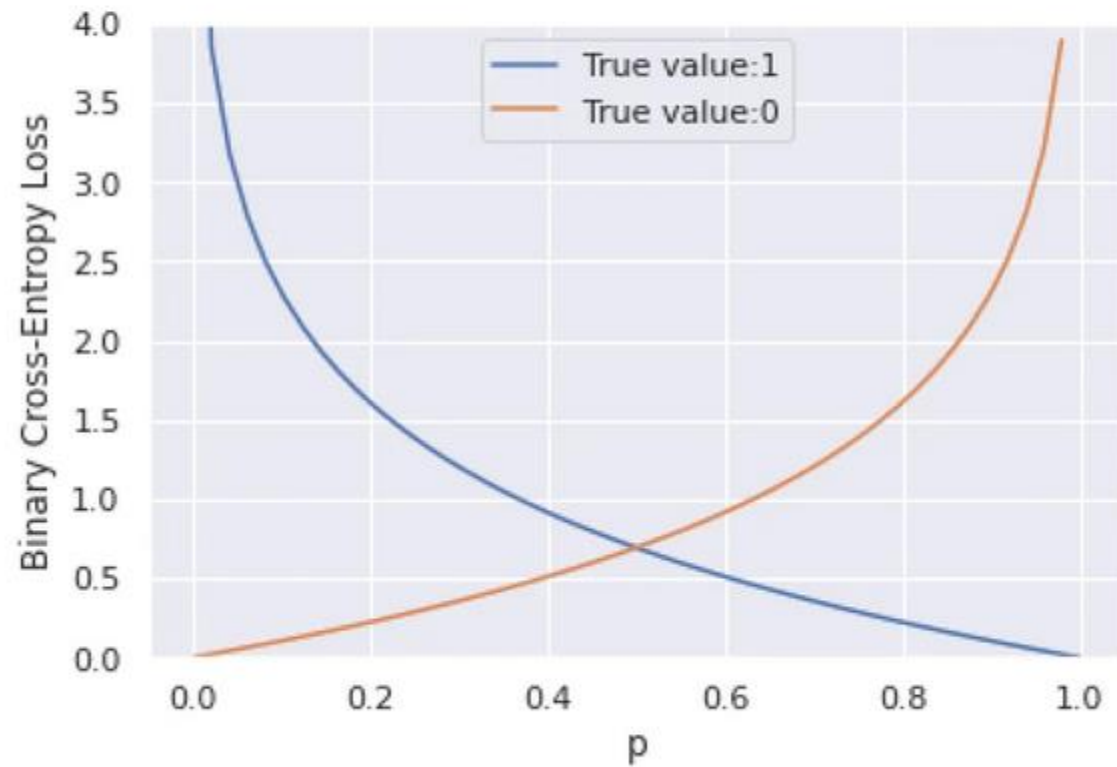
$$E = -d \log(y) - (1-d) \log(1-y)$$



Cross-Entropy

$$\text{BinaryCrossEntropy} = -\frac{1}{N} \sum_{i=1}^N [y_i * \log(y_{pred}) + (1 - y_i) * \log(1 - y_{pred})]$$

Binary cross entropy formula



<https://www.v7labs.com/blog/cross-entropy-loss-guide>

$$\min_w E$$

$$w_j \leftarrow w_j - \eta \frac{\partial E}{\partial w_j}$$

$$\frac{\partial E}{\partial w_j} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j}$$

$$E = -d \log(y) - (1-d) \log(1-y)$$

$$\min_w E$$

$$w_j \leftarrow w_j - \eta \frac{\partial E}{\partial w_j}$$

$$\frac{\partial E}{\partial w_j} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j}$$

$$E = -d \log(y) - (1-d) \log(1-y)$$

$$\frac{\partial E}{\partial y} = -\frac{d}{y} + \frac{1-d}{1-y} = -\frac{d-y}{y(1-y)}$$

$$\frac{\partial E}{\partial w_j} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j}$$

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$$\frac{\partial E}{\partial w_j} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j}$$

$$= -\frac{(d-y)}{y(1-y)} \sigma'(v) x_j$$

$$= -(d-y)x_j \quad !$$

$$\min_w E$$

$$w_j \leftarrow w_j - \eta \frac{\partial E}{\partial w_j}$$

$$\frac{\partial E}{\partial w_j} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j}$$

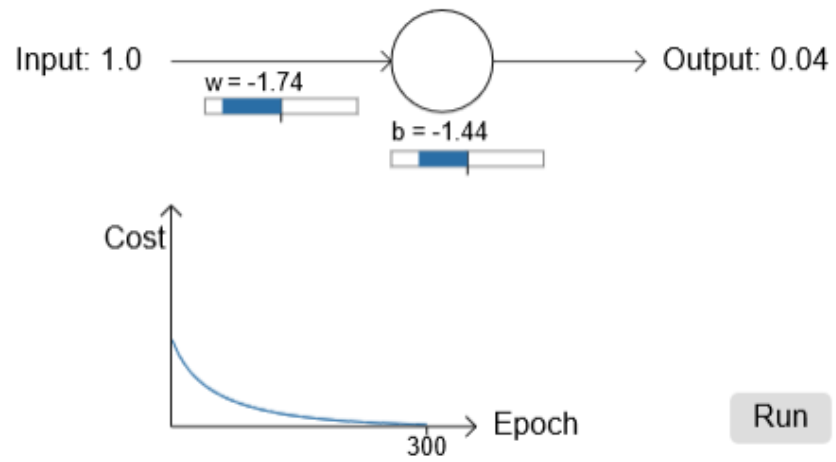
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$$\begin{aligned} \frac{\partial E}{\partial w_j} &= \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j} \\ &= -\frac{(d-y)}{y(1-y)} \sigma'(v) x_j \end{aligned}$$

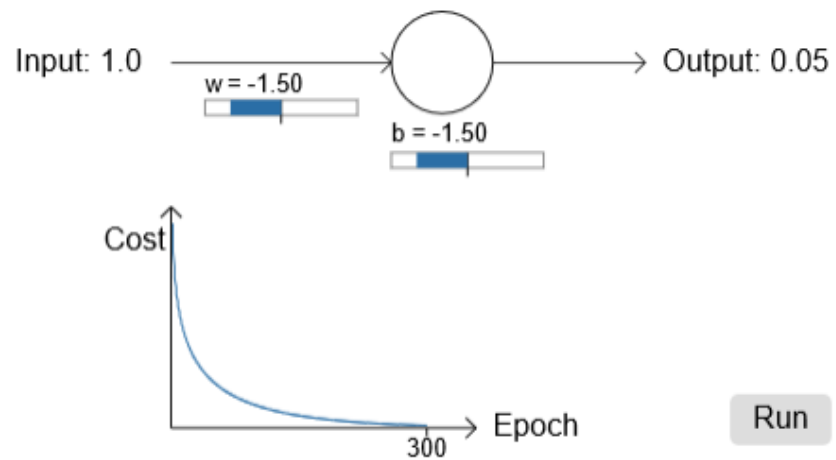
$$= -(d-y)x_j \quad !$$

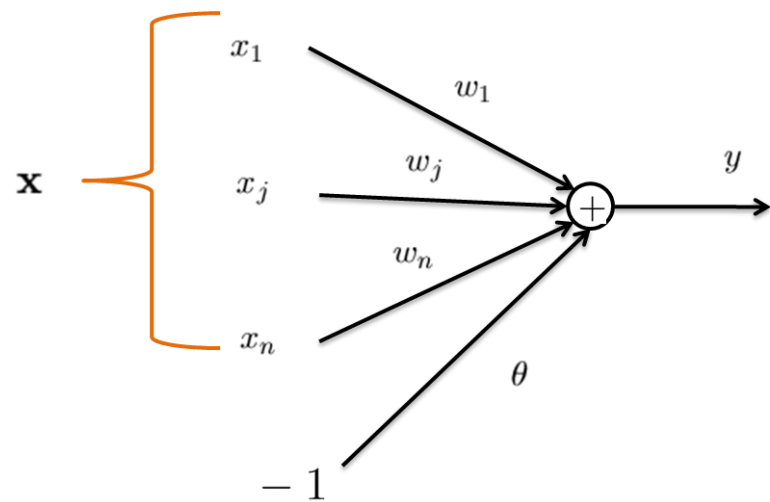
$$w_j \leftarrow w_j + \eta(d-y)x_j$$



Insurprisingly, the neuron learns perfectly well in this instance, just as it did earlier. And now let's look at the case where our neuron got stuck before ([link](#), for comparison), with the weight and bias both starting at 2.0:

<http://neuralnetworksanddeeplearning.com/chap3.html>



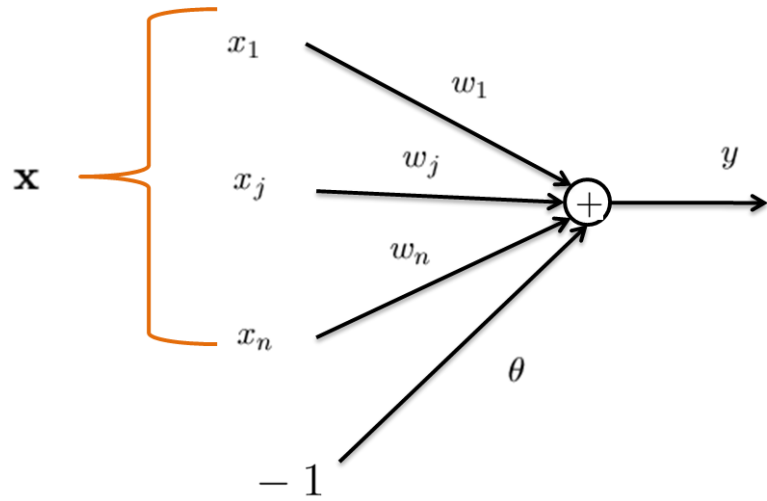


$$\mathbf{x} \rightarrow d \quad \longrightarrow \quad E = 0.5(d - y)^2$$

$$\frac{\partial E}{\partial w_j} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j}$$

$$= -(d - y)x_j$$

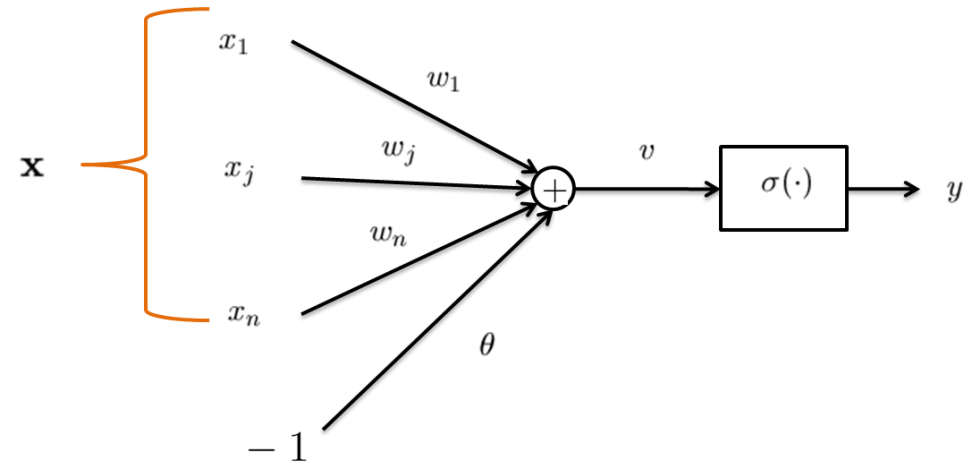
$$w_j \leftarrow w_j + \eta(d - y)x_j$$



$$\mathbf{x} \rightarrow d \quad \longrightarrow \quad E = 0.5(d - y)^2$$

$$\begin{aligned} \frac{\partial E}{\partial w_j} &= \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j} \\ &= -(d - y)x_j \end{aligned}$$

$$w_j \leftarrow w_j + \eta(d - y)x_j$$



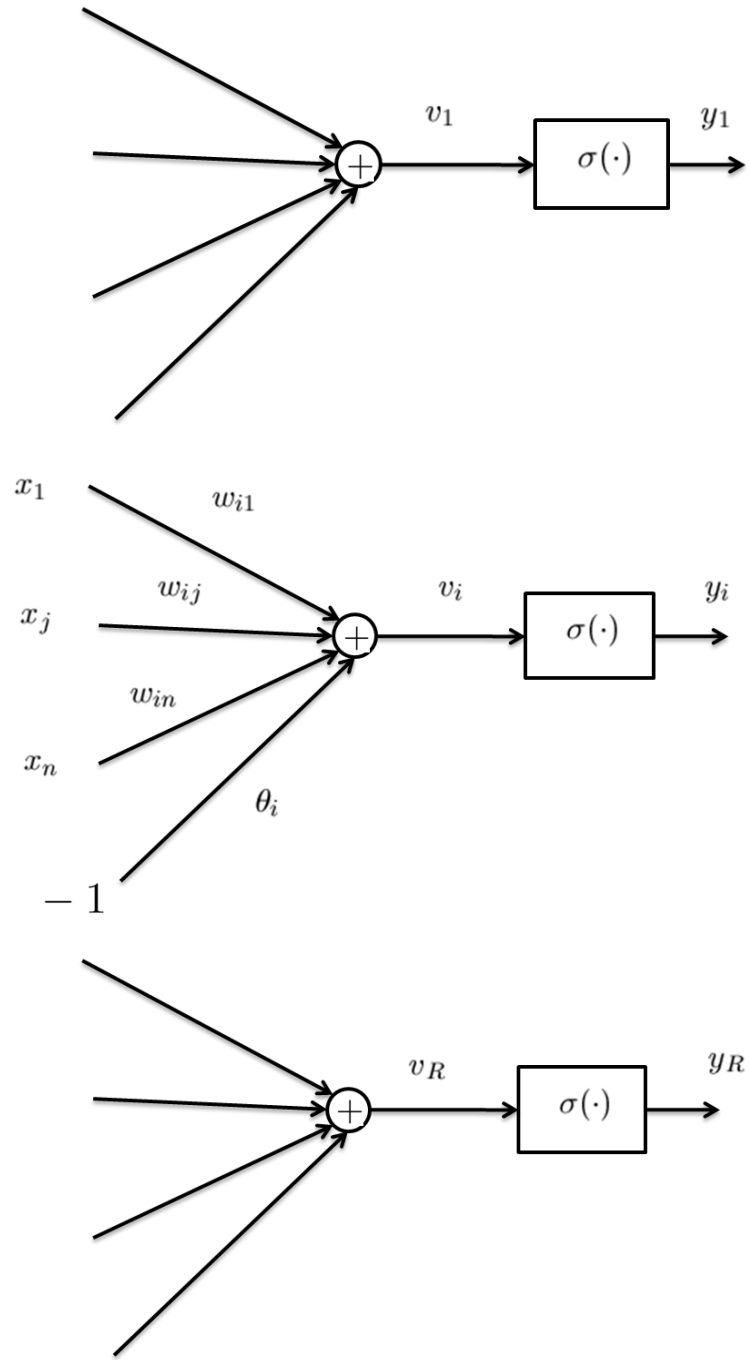
$$E = -d \log(y) - (1-d) \log(1-y)$$

$$\begin{aligned} \frac{\partial E}{\partial w_j} &= \frac{\partial E}{\partial y} \frac{\partial y}{\partial v} \frac{\partial v}{\partial w_j} \\ &= -\frac{(d-y)}{y(1-y)} \sigma'(v) x_j \end{aligned}$$

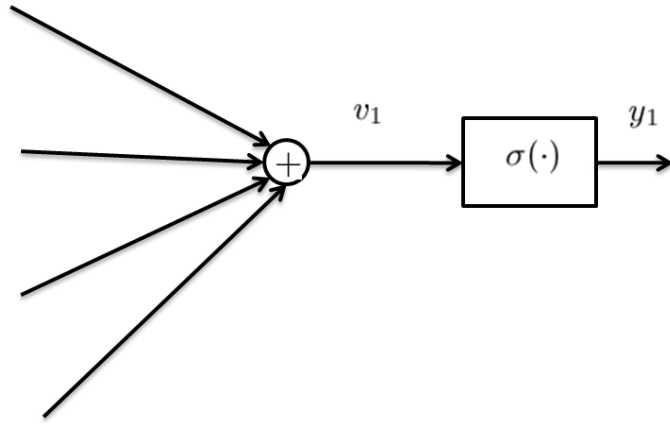
$$\boxed{= -(d - y)x_j} \quad !$$

$$w_j \leftarrow w_j + \eta(d - y)x_j$$

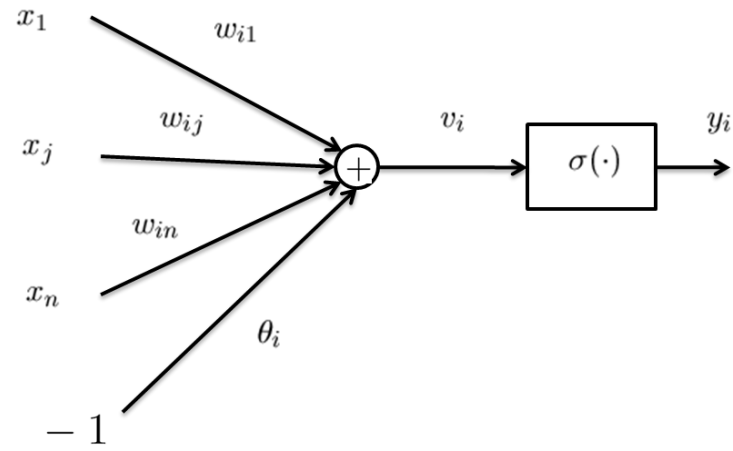
R class classification (Multicategory case)



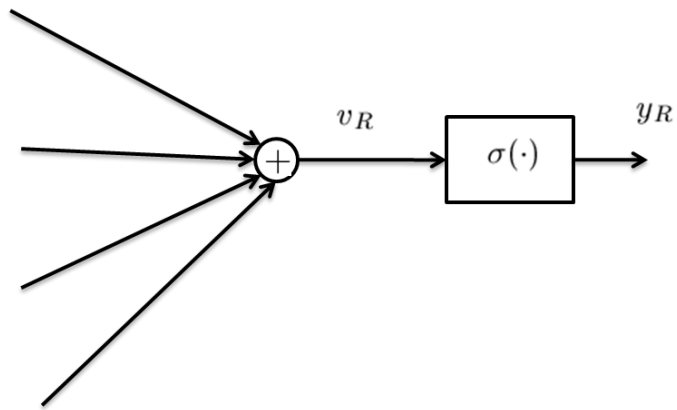
R class classification (Multicategory case)



Class 1 $\rightarrow d_1$

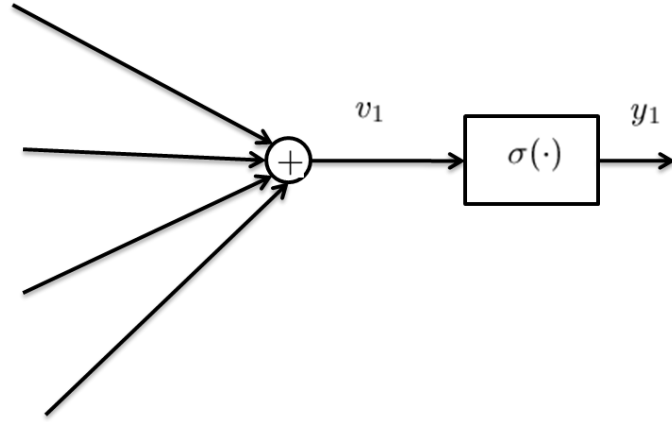


Class i $\rightarrow d_i$



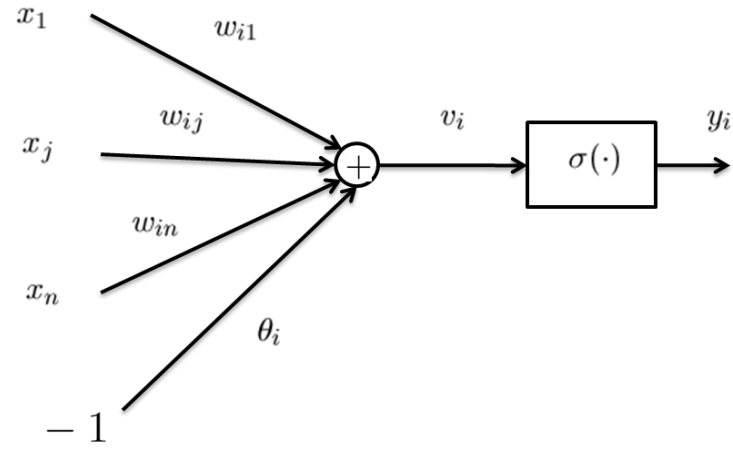
Class R $\rightarrow d_R$

R class classification (Multicategory case)

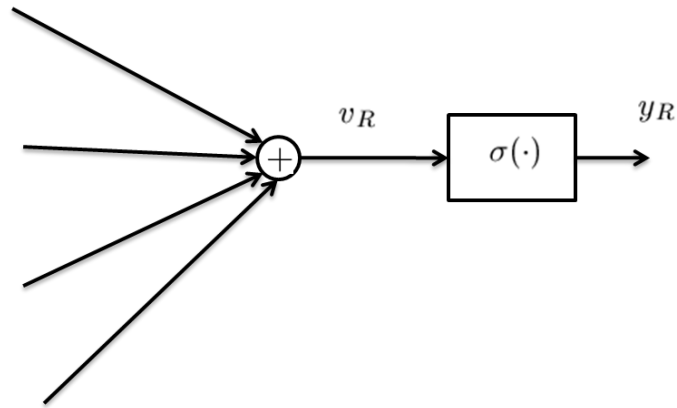


Class 1 $\rightarrow d_1$

$$E = - \sum_{k=1}^R [d_k \log(y_k) + (1 - d_k) \log(1 - y_k)]$$

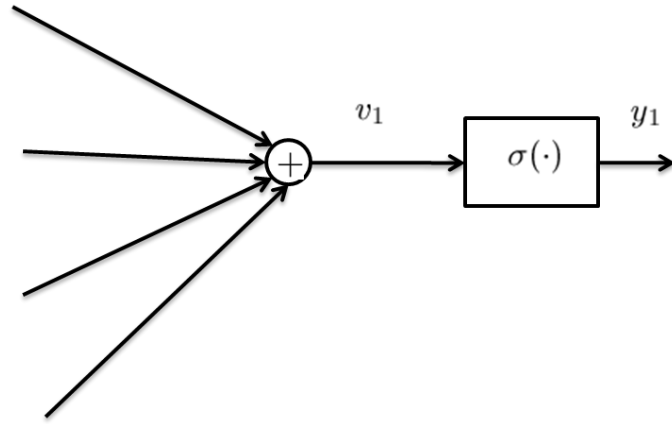


Class i $\rightarrow d_i$



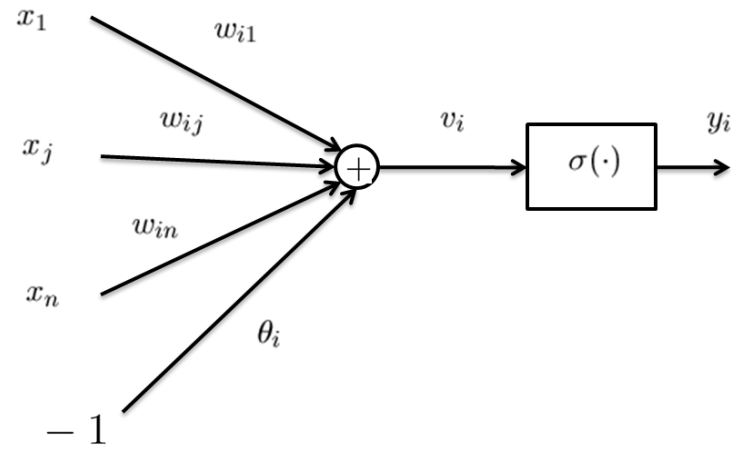
Class R $\rightarrow d_R$

R class classification (Multicategory case)



Class 1 $\rightarrow d_1$

$$E = - \sum_{k=1}^R [d_k \log(y_k) + (1 - d_k) \log(1 - y_k)]$$

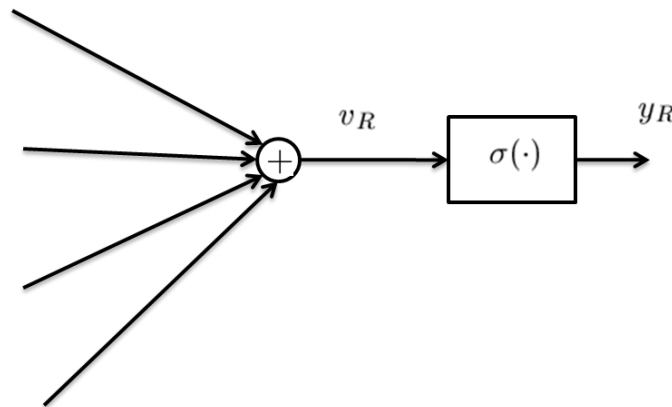


Class i $\rightarrow d_i$

$$\frac{\partial E}{\partial w_{ij}} = \frac{\partial E}{\partial y_i} \frac{\partial y_i}{\partial v_i} \frac{\partial v_i}{\partial w_{ij}}$$

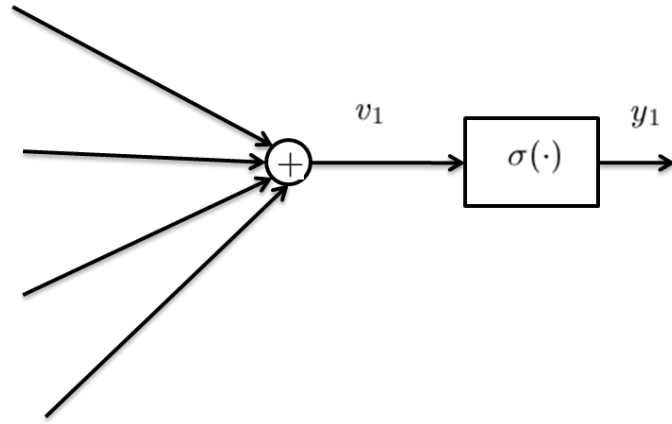
$$= - \frac{(d_i - y_i)}{y_i(1 - y_i)} \sigma'(v_i) x_j$$

$$= -(d_i - y_i) x_j$$



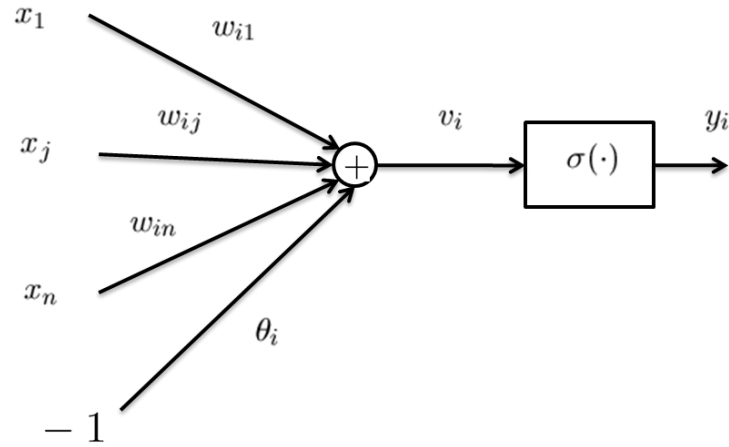
Class R $\rightarrow d_R$

R class classification (Multicategory case)



Class 1 $\rightarrow d_1$

$$E = - \sum_{k=1}^R [d_k \log(y_k) + (1 - d_k) \log(1 - y_k)]$$

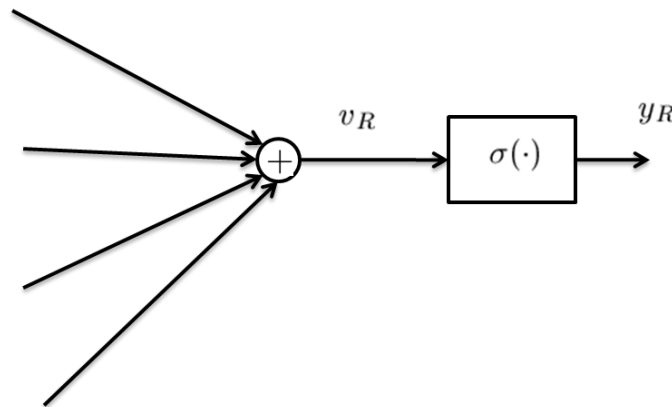


Class i $\rightarrow d_i$

$$\frac{\partial E}{\partial w_{ij}} = \frac{\partial E}{\partial y_i} \frac{\partial y_i}{\partial v_i} \frac{\partial v_i}{\partial w_{ij}}$$

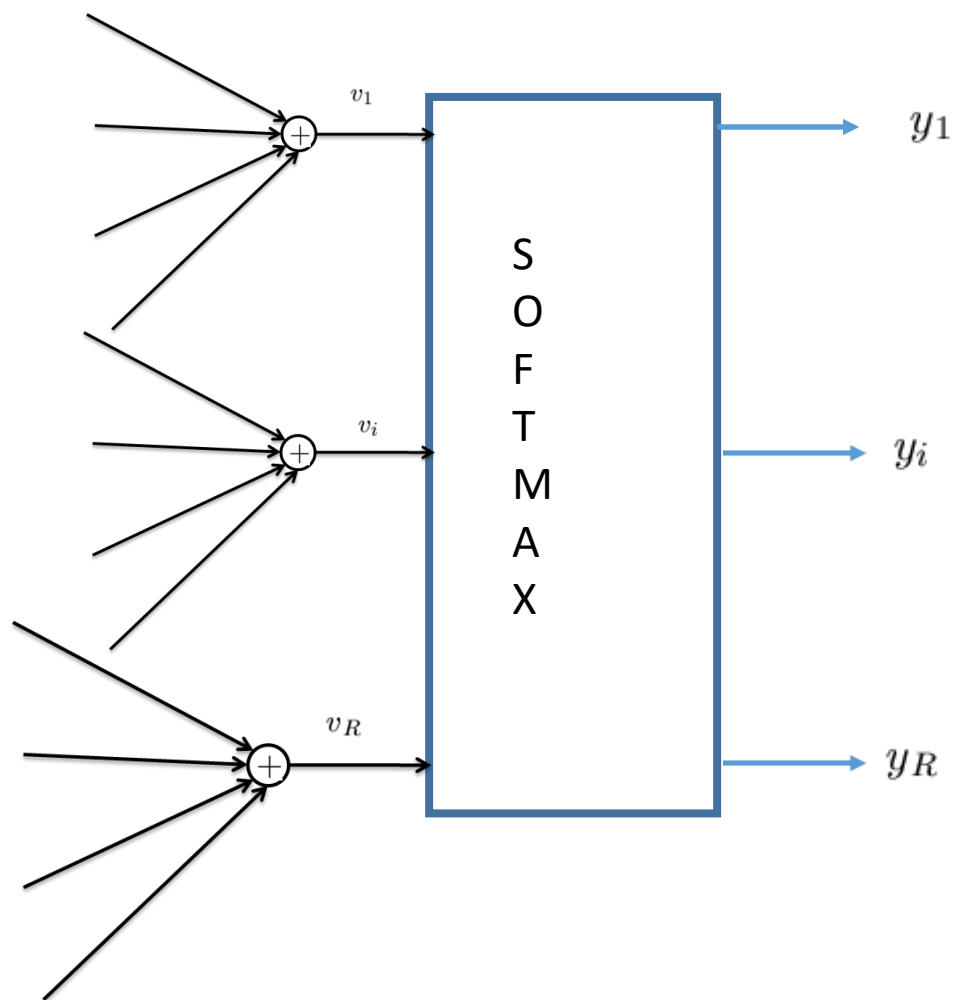
$$= - \frac{(d_i - y_i)}{y_i(1 - y_i)} \sigma'(v_i) x_j$$

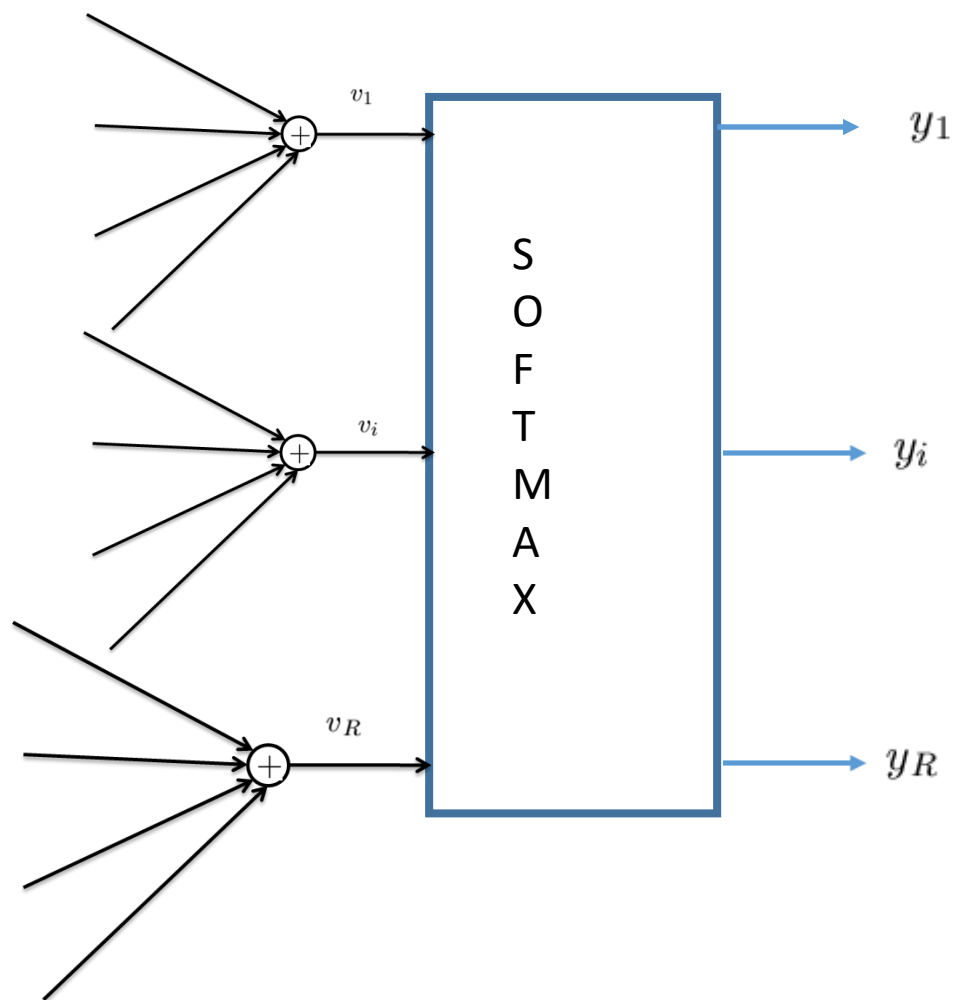
$$= -(d_i - y_i) x_j$$



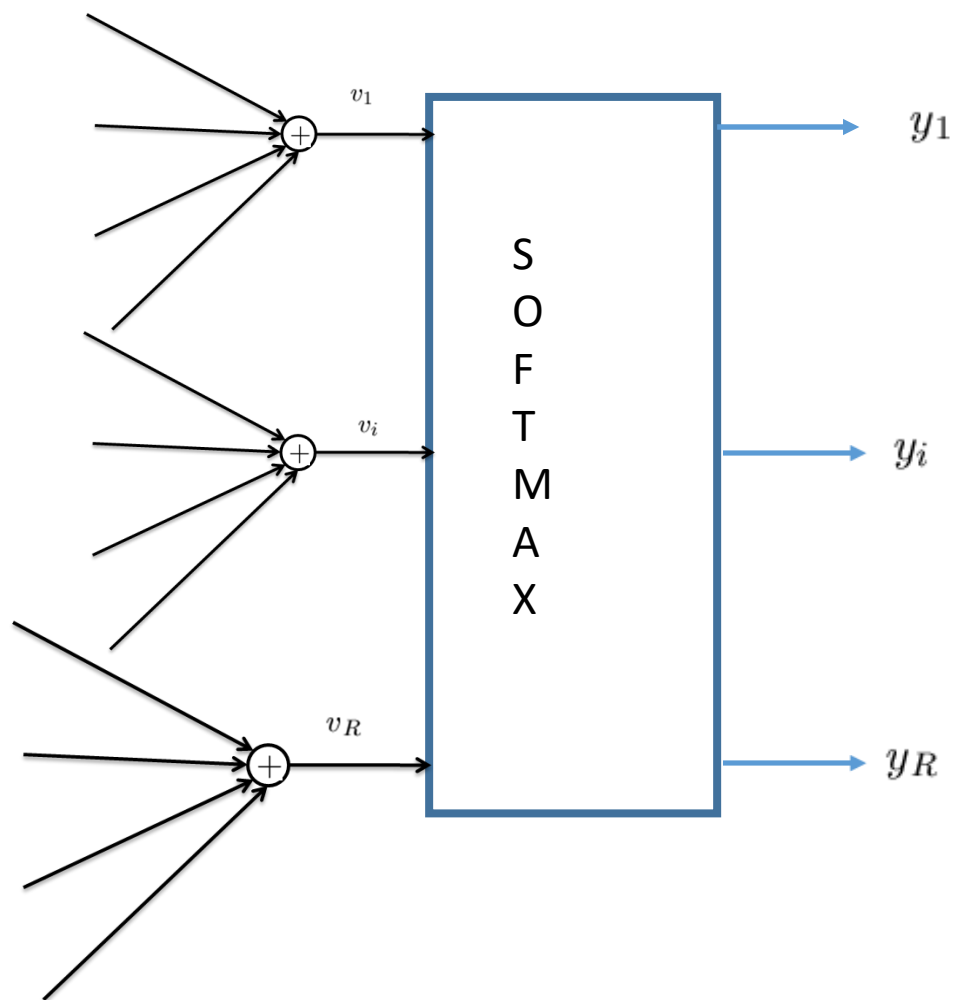
Class R $\rightarrow d_R$

$$w_{ij} \leftarrow w_{ij} + \eta (d_i - y_i) x_j$$





$$y_i = \frac{e^{v_i}}{e^{v_1} + \dots + e^{v_i} + \dots + e^{v_R}}$$



$$y_i = \frac{e^{v_i}}{e^{v_1} + \dots + e^{v_i} + \dots + e^{v_R}}$$

$$0 \leq y_i \leq 1$$

$$\sum_{i=1}^R y_i = 1$$

$$\frac{dy_i}{dv_i} = \frac{\sum_{k \neq i} e^{v_i} e^{v_k}}{(\sum_i e^{v_i})^2} \geq 0$$

$$y_i = \frac{e^{v_i}}{e^{v_1} + \dots + e^{v_i} + \dots + e^{v_R}}$$

$$\frac{dy_k}{dv_i} = - \frac{e^{v_i} e^{v_k}}{(\sum_i e^{v_i})^2} \leq 0 \quad k \neq i$$

$$0 \leq y_i \leq 1$$

$$\sum_{i=1}^R y_i = 1$$

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$$y_i = \frac{e^{v_i}}{e^{v_1} + \dots + e^{v_i} + \dots + e^{v_R}}$$

$$\frac{dy_k}{dv_i} = - \frac{e^{v_i} e^{v_k}}{(\sum_i e^{v_i})^2} \leq 0 \quad k \neq i$$

$$0 \leq y_i \leq 1$$

$$\sum_{i=1}^R y_i = 1$$



$z_1^L =$

3.6



$z_2^L =$

0.2



$z_3^L =$

0



$z_4^L =$

0.7



$a_1^L =$

0.896



$a_2^L =$

0.030



$a_3^L =$

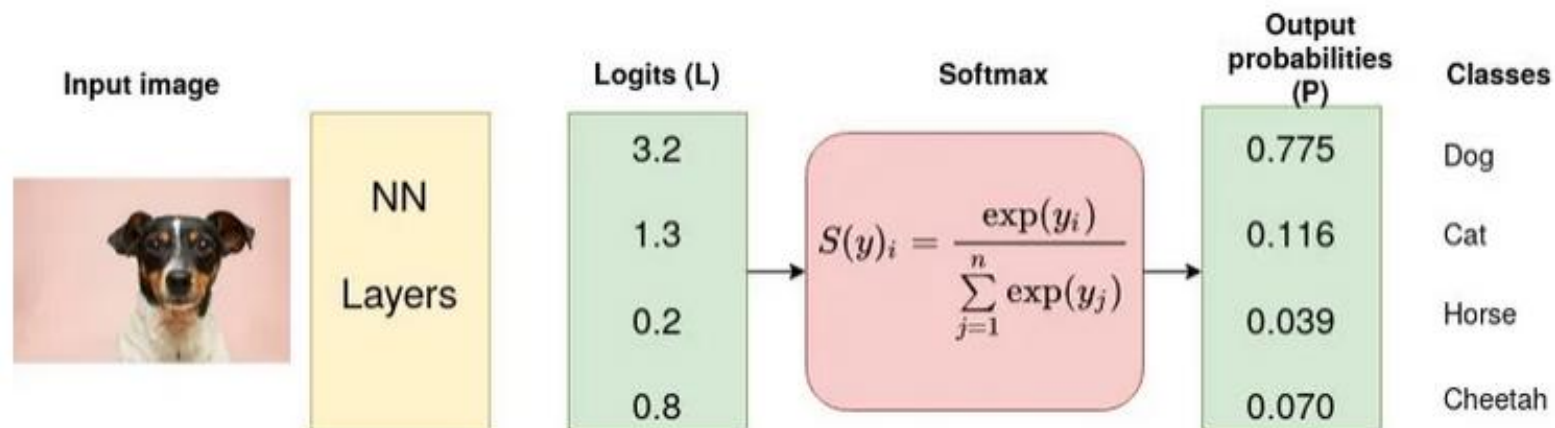
0.024



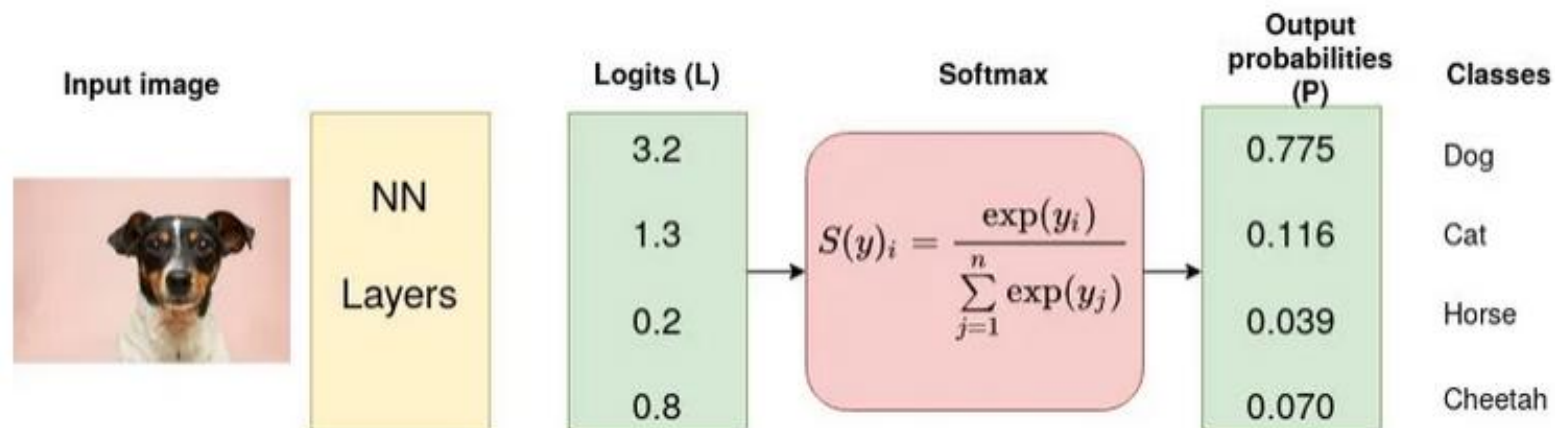
$a_4^L =$

0.049

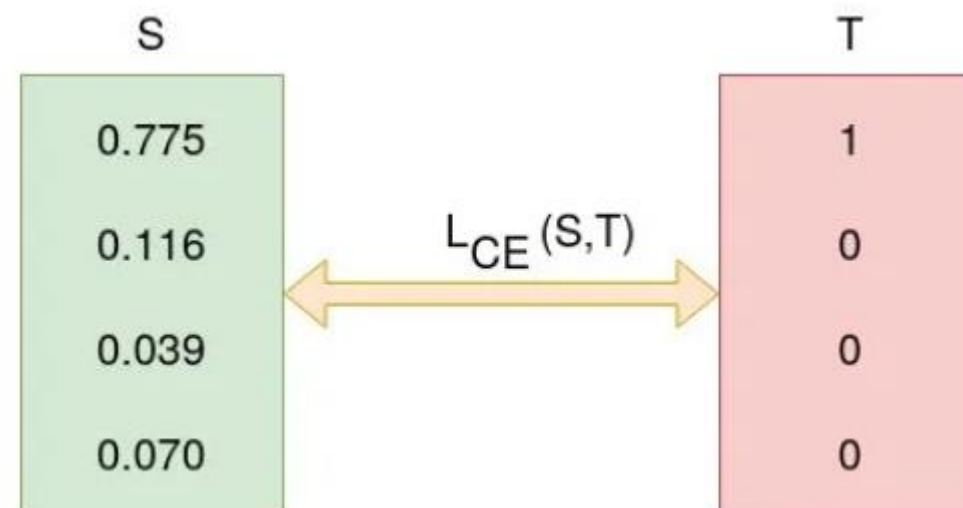
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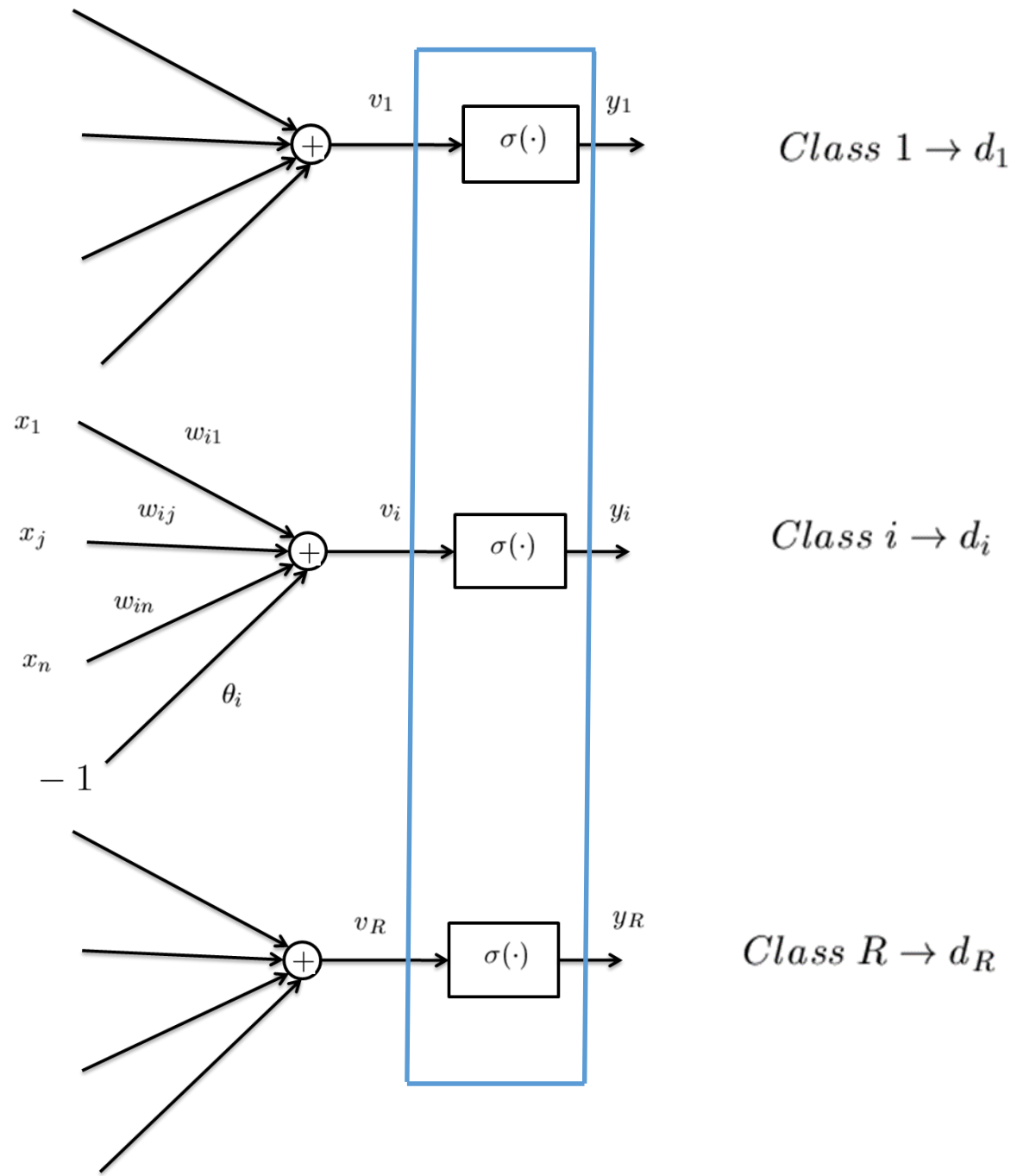


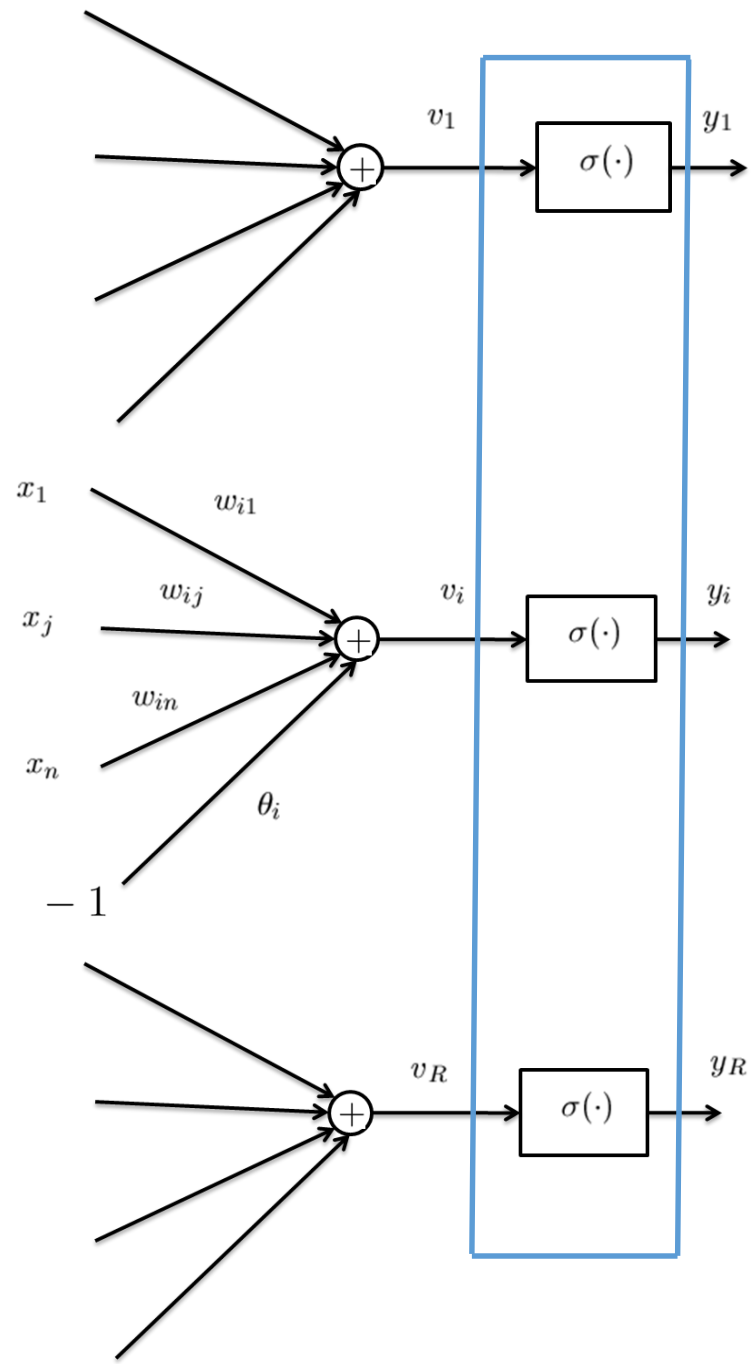
Input image source: Photo by [Victor Grabarczyk](#) on [Unsplash](#) . Diagram by author.



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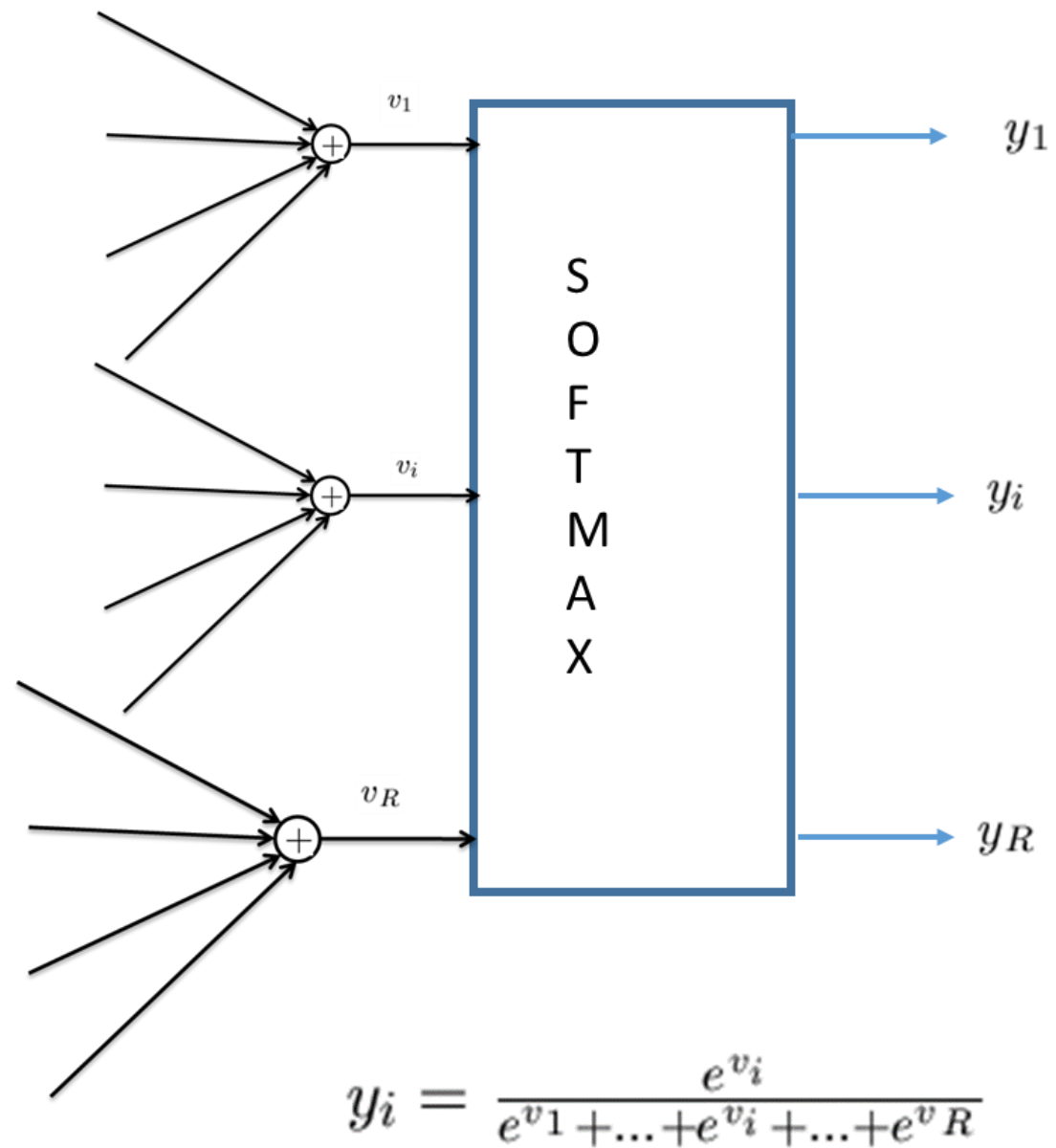




Class 1 $\rightarrow d_1$

Class i $\rightarrow d_i$

Class R $\rightarrow d_R$



$x \in \text{Class } k$



$$d_k = \begin{pmatrix} d_{k1} \\ d_{k2} \\ \vdots \\ d_{kk} \\ \vdots \\ d_{kn} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$



One-Hot-Vector

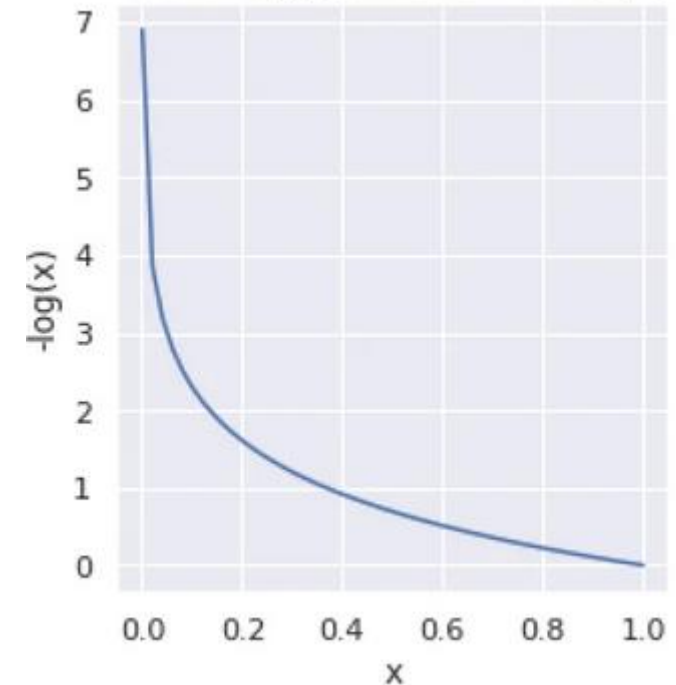
$$E_x = - \sum_{j=1}^R d_{kj} \log(y_k) \rightarrow -\log(y_k)$$

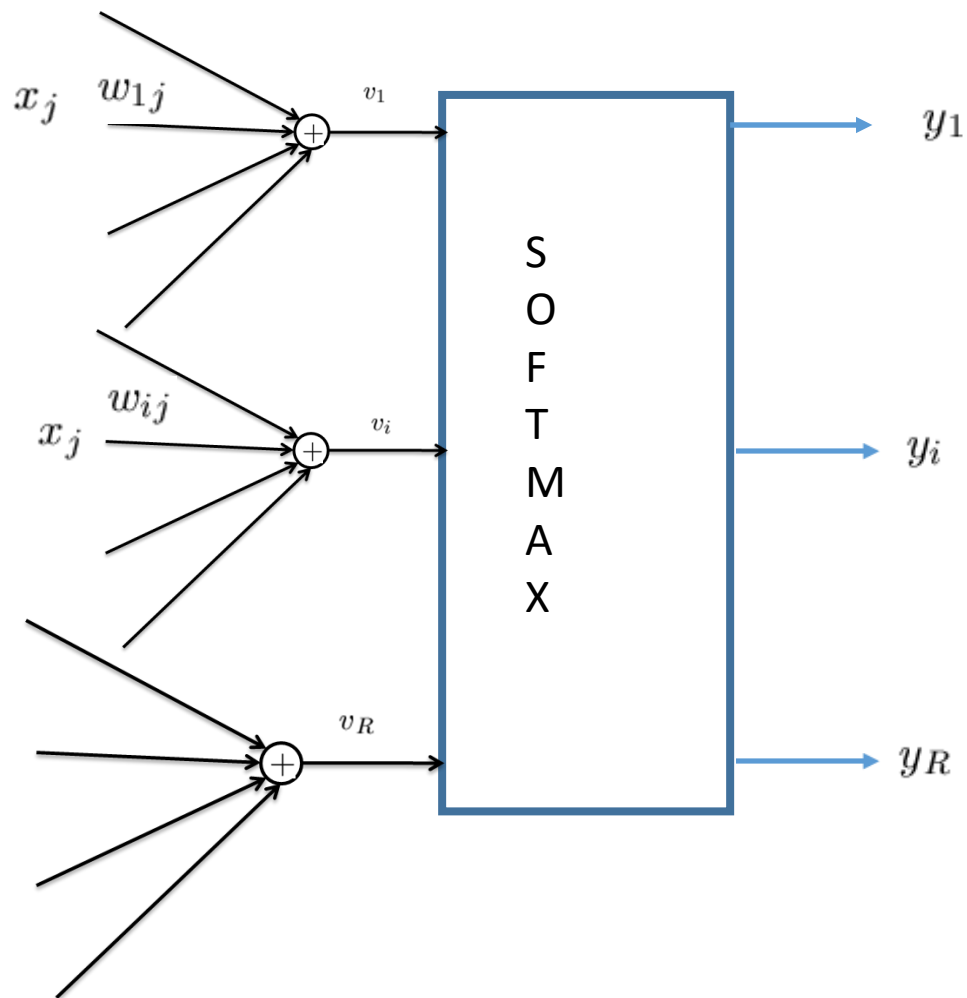


Categorical Cross Entropy
Softmax Loss
Log-likelihood cost

$$E = \frac{1}{N} \sum_x E_x$$

Plot of $-\log(x)$ in the interval $(0,1]$





$$y_i = \frac{e^{v_i}}{e^{v_1} + \dots + e^{v_i} + \dots + e^{v_R}}$$

Assume that $x \in \text{Class } i \rightarrow E = -\log y_i$

$$w_{1j} \leftarrow w_{1j} - \eta \frac{\partial E}{\partial w_{1j}}$$

$$w_{ij} \leftarrow w_{ij} - \eta \frac{\partial E}{\partial w_{ij}}$$

$$\frac{\partial E}{\partial w_{1j}} = \frac{\partial E}{\partial y_i} \frac{\partial y_i}{\partial v_1} \frac{\partial v_1}{\partial w_{1j}}$$

$$\frac{\partial E}{\partial y_i} = -\frac{1}{y_i}$$

$$\frac{\partial y_i}{\partial v_1} = -\frac{e^{v_1} e^{v_i}}{(\sum_j e^{v_j})^2} = -y_i y_1$$

$$\frac{\partial E}{\partial w_{1j}} = \frac{1}{y_i} y_i y_1 x_j = y_1 x_j$$

$$w_{1j} \leftarrow w_{1j} - \eta y_1 x_j$$

$$y_i = \frac{e^{v_i}}{e^{v_1} + \dots + e^{v_i} + \dots + e^{v_R}}$$

$$\frac{\partial E}{\partial w_{1j}} = \frac{\partial E}{\partial y_i} \frac{\partial y_i}{\partial v_1} \frac{\partial v_1}{\partial w_{1j}}$$

$$\frac{\partial E}{\partial y_i} = -\frac{1}{y_i}$$

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$$\frac{\partial E}{\partial w_{1j}} = \frac{1}{y_i} y_i y_1 x_j = y_1 x_j$$

$$w_{1j} \leftarrow w_{1j} - \eta y_1 x_j$$

$$y_i = \frac{e^{v_i}}{e^{v_1} + \dots + e^{v_i} + \dots + e^{v_R}}$$

$$\frac{\partial E}{\partial w_{ij}} = \frac{\partial E}{\partial y_i} \frac{\partial y_i}{\partial v_i} \frac{\partial v_i}{\partial w_{ij}}$$

$$\frac{\partial E}{\partial y_i} = -\frac{1}{y_i}$$

$$\frac{\partial y_i}{\partial v_i} = \frac{e^{v_i} (\sum_j e^{v_j}) - e^{v_i} e^{v_i}}{(\sum_j e^{v_j})^2} = y_i - y_i^2$$

$$\frac{\partial E}{\partial w_{1j}} = -\frac{1}{y_i} y_i (1 - y_i) x_j = -(1 - y_i) x_j$$

$$w_{1j} \leftarrow w_{1j} + \eta (1 - y_i) x_j$$